Idiopathic Neonatal Hepatitis and Giant Cell Hepatitis: A Shrinking Diagnosis, Giant Cell Hepatitis with Autoimmune Hemolytic Anemia, and Prognosis.

Picture an eight-week-old boy referred for jaundice. His direct bilirubin, the conjugated fraction that the liver has processed but cannot excrete, is four milligrams per deciliter. His stools are yellow, not pale. The ultrasound shows a gallbladder and no cyst. A liver biopsy is done, and the report comes back with a phrase that sounds like an answer: neonatal giant cell hepatitis. The chart is updated to say idiopathic neonatal hepatitis, and everyone relaxes.

Now picture a nine-month-old girl brought in for pallor. Her hemoglobin is six grams per deciliter, and the direct antiglobulin test, often called the Coombs test, is positive for both immunoglobulin G and complement. She is treated for autoimmune hemolytic anemia. Two weeks later someone checks her liver enzymes, and her alanine aminotransferase is above one thousand eight hundred units per liter. She too has giant cells on her liver biopsy.

These two children share a histological pattern and almost nothing else. The clinical question of this chapter is what idiopathic neonatal hepatitis really means today, what giant cells on a liver biopsy do and do not tell you, how to recognize the distinct disease called giant cell hepatitis with autoimmune hemolytic anemia, and what prognosis to give a family. The main idea is simple. Idiopathic neonatal hepatitis is a provisional label for a list of tests not yet done. You earn it by excluding biliary atresia, then treatable causes, then genetic disease, and you re-examine it at every visit. Giant cells are a pattern, not a diagnosis. And giant cell hepatitis with autoimmune hemolytic anemia is a separate, antibody-driven and complement-mediated disease of older infants that needs early, aggressive immunosuppression.

A note on sources. For the cholestatic infant we use the 2017 joint guideline on cholestatic jaundice in infants, written by Fawaz and colleagues for the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, known as NASPGHAN, and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, known as ESPGHAN. For how common the idiopathic label has been, we use a 2015 systematic review by Gottesman and colleagues, the Bergamo cohort reported by Nicastro and colleagues in 2019, a Japanese gene panel study by Togawa and colleagues in 2016, and a Chinese single center comparison by Wang and colleagues in 2022. For histology we use a pathology series by Torbenson and colleagues in 2010 and the biopsy scoring study from the Childhood Liver Disease Research Network by Russo and colleagues in 2016. For classic prognosis we use three Archives of Disease in Childhood papers, by Danks in 1977, by Odievre in 1981, and by Deutsch in 1985. For giant cell hepatitis with autoimmune hemolytic anemia we use the original description by Bernard in 1981, the long-term cohort by Maggiore and colleagues in 2011, the complement study by Whitington and colleagues in 2014, the reviews by Nastasio

and colleagues in 2021 and by Poddighe and colleagues in 2021, and the Shanghai rituximab cohort by Zhang and colleagues in 2024.

Part one. Four names that are not synonyms.

Four terms are often used as if they meant the same thing, and they do not. Neonatal hepatitis is purely descriptive. It means intrahepatic cholestasis with evidence of injury to hepatocytes, the main liver cells, and it says nothing about cause. Idiopathic neonatal hepatitis, which we will abbreviate as INH, means that no cause has been found after a complete evaluation, including exclusion of biliary atresia, infection, endocrine disease, metabolic disease and genetic disease. It is a residue: whatever is left over when the tests run out. Neonatal giant cell hepatitis is a histological description. It means that the pathologist sees large hepatocytes containing many nuclei, called multinucleated giant cells, usually with disarray of the liver cell plates. And giant cell hepatitis with autoimmune hemolytic anemia, which we will simply call the anemia variant, is a specific clinical syndrome. It combines giant cell hepatitis with a Coombs-positive hemolytic anemia, it begins between about one month and two years of age, and it was first described by Bernard and colleagues in Paris in 1981.

Why does the distinction matter? Because an infant can carry the histological label of giant cell hepatitis and still have Alagille syndrome, hypopituitarism or even biliary atresia. A pathology report that says giant cell hepatitis is never the final diagnosis. And the anemia variant, despite its name, is not really a neonatal disease at all. It begins after the newborn period, it needs immunosuppression, and it behaves like an antibody-mediated disease. The anemia variant shares a pattern with neonatal hepatitis, not a mechanism.

Part two. The shrinking category.

For decades, idiopathic neonatal hepatitis was one of the two largest groups in any series of cholestatic infants. The clearest summary is the 2015 systematic review by Gottesman and colleagues, which pooled seventeen studies and one thousand six hundred and ninety-two infants with conjugated hyperbilirubinemia. Idiopathic hepatitis was the single largest category, at twenty-six point zero percent. Biliary atresia was a close second at twenty-five point nine percent. Infection accounted for about eleven and a half percent, parenteral nutrition for about six percent, metabolic disease for about four percent, and alpha-one antitrypsin deficiency for about four percent. Further down the list were perinatal hypoxia, paucity of bile ducts, choledochal cyst and endocrine disease. Most striking were the genetic cholestatic syndromes at the bottom. Progressive familial intrahepatic cholestasis, known as PFIC, was only one percent, and Alagille syndrome was just under one percent.

Those last two numbers are the key to reading the whole chart. They are far lower than what modern genetic series find, which tells us that many infants with PFIC and Alagille syndrome were sitting inside the twenty-six percent called idiopathic. The authors themselves warned

that the idiopathic group was defined loosely and probably overestimated true idiopathic disease. So a pooled idiopathic share measures the diagnostic tools of an era, not biology.

Modern series show the residue shrinking. In Bergamo, Nicastro and colleagues enrolled one hundred and twenty-five infants with cholestasis under one year of age between 2012 and 2017 and applied an algorithm that included next-generation sequencing. Biliary atresia made up fifty-nine percent. Fifty children had genetic testing, by single gene, virtual panel or trio exome, and sixty percent of those tested received a diagnosis, most often Alagille syndrome, PFIC type two, alpha-one antitrypsin deficiency and PFIC type three. In the end, only fourteen percent of the whole cohort remained without a cause. A Chinese single center saw the same trend over time: unexplained cholestasis fell from about nine percent of infants in a 2012 series to two and a half percent in a 2019 cohort, with genetic testing adding about five and a half percent of diagnoses.

The 2017 joint guideline puts it in words. It calls idiopathic neonatal hepatitis a descriptive term whose frequency continues to decline as diagnostic evaluation improves. It estimates that biliary atresia accounts for twenty-five to forty percent of infantile cholestasis and that a collection of individually rare genetic disorders accounts for about twenty-five percent. And it advises a low threshold for gene panels or exome sequencing.

Part three. What genetic testing finds.

The yield of genetic testing depends heavily on who is tested. In the Japanese study by Togawa and colleagues, one hundred and nine infants with intrahepatic cholestasis were tested with an eighteen-gene panel, and twenty-six percent received a molecular diagnosis. When the infants were divided by clinical picture, the yield was seventy-one percent in those whose phenotype already suggested a genetic cholestasis, but it was still twelve and a half percent, roughly one in eight, in infants classified as unknown cause without complications. The diagnoses included Alagille syndrome, the neonatal form of Dubin-Johnson syndrome, citrin deficiency, and low gamma-glutamyl transferase PFIC. In a large North American commercial laboratory series reported by Karpen and colleagues in 2021, a sixty-six-gene panel in more than two thousand referred patients gave a yield of about twelve percent, and a Japanese study by Ito and colleagues in 2022 found about twenty-seven percent with a sixty-one-gene panel.

So what is hiding inside the idiopathic group? The usual answers are Alagille syndrome from variants in JAG1 or NOTCH2; PFIC type two from the bile salt export pump gene, ABCB11; tight junction protein two deficiency; citrin deficiency; the inborn errors of bile acid synthesis; the arthrogryposis, renal dysfunction and cholestasis syndrome; alpha-one antitrypsin deficiency; and hypopituitarism, which is not genetic in most cases but is easily missed. The practical lesson is that yield is highest in infants with a suggestive phenotype, such as low gamma-glutamyl transferase despite cholestasis, dysmorphic features, a family history or consanguinity, but that even an unremarkable infant labeled idiopathic has about a one in eight

chance of a genetic answer.

Part four. Giant cells are a pattern, not a diagnosis.

Torbenson and colleagues reviewed sixty-two neonatal liver biopsies that showed giant cell hepatitis and traced the final diagnosis in each. Forty-nine percent remained idiopathic. Fifteen percent had hypopituitarism. Eight percent had biliary atresia. About six percent had Alagille syndrome, another six percent had PFIC or a bile salt defect, five percent had neonatal hemochromatosis, four percent had a viral infection, and eight percent had other causes. In other words, even in a pathology series, about half of the infants with this pattern had a specific cause, and one in seven had a hormone deficiency that is entirely treatable. When the biopsy shows giant cells, check cortisol, thyroid stimulating hormone and free thyroxine and blood glucose, and look for micropenis and midline defects.

Why does the newborn liver produce giant cells at all? The honest answer is that the exact mechanism is unproven. The leading idea is that neonatal hepatocytes are still proliferating, with an immature bile acid pool and immature transporters, and that when they are injured, whether by cholestasis and toxic bile acids, by missing hormones such as cortisol and thyroxine, by a virus, or by antibody and complement, they fuse with each other or fail to complete cell division and form multinucleated cells. Different causes converge on the same histology, so the pattern cannot name the cause. The tendency fades with age, and giant cell hepatitis after infancy is rare and points to viruses, drugs or autoimmunity. The cause is therefore found with clinical chemistry, hormones, imaging and genes, not with the giant cells themselves.

Part five. Can the biopsy separate hepatitis from atresia?

The most dangerous misreading in this field is to accept a hepatitis pattern in an infant who actually has biliary atresia. The 2017 guideline notes that giant cell transformation occurs in twenty to fifty percent of biopsies from infants with biliary atresia, although it is usually less prominent than in idiopathic hepatitis. The best data on biopsy performance come from Russo and colleagues, who scored two hundred and twenty-seven needle biopsies from cholestatic infants in the Childhood Liver Disease Research Network using a consensus set of features. For the diagnosis of biliary atresia, accuracy was ninety point one percent, with a ninety-five percent confidence interval from about eighty-five to ninety-five percent. Sensitivity was eighty-eight point four percent, which means that about one infant with atresia in nine was read as non-obstructive. Specificity was ninety-two point seven percent. The three features that best predicted atresia were bile plugs in portal ducts or ductules, a moderate to marked ductular reaction, and portal stromal edema. The guideline quotes ninety to ninety-five percent accuracy when the biopsy is read by an experienced pediatric liver pathologist.

What favors hepatitis is lobular disarray, prominent giant cell change, extramedullary hematopoiesis, and portal tracts without ductular reaction or bile plugs. But here is the trap: a biopsy taken in the first weeks of life can lack the obstructive features that will appear later. If the stools remain pale, the stool color overrules the histology. Repeat the biopsy or proceed to cholangiography, and never let a report of neonatal hepatitis delay a Kasai operation.

Part six. The classic prognosis.

Most of what we know about the natural history of idiopathic neonatal hepatitis comes from series published before genetic testing existed, and they need to be read with that in mind. In Paris, Odievre and colleagues followed one hundred and three infants with cholestasis that began before three months of age and a patent biliary tree. Eighty-one, about seventy-nine percent, recovered within months. Twenty-two developed progressive chronic liver disease, and the poor outcomes clustered in alpha-one antitrypsin deficiency, paucity of interlobular ducts, and familial idiopathic hepatitis.

In Melbourne, Danks and colleagues followed one hundred and five babies for up to eleven and a half years and found that stool color predicted outcome. Among infants with persistently acholic, that is pale, stools, about forty percent died or developed cirrhosis. Among those with a less obstructive picture, fewer than fifteen percent did. Deutsch and colleagues later followed the Melbourne group for six to eighteen years. Among sporadic cases, only three of forty reviewed patients developed chronic liver disease, about seven and a half percent, although many had other permanent disabilities. Familial cases were very different. Four of nine infants with familial idiopathic hepatitis died within twelve months, and two of the four survivors who were reviewed also died later. Their disease progressed continuously, without the period of recovery seen in sporadic cases.

How should we read these numbers now? The familial idiopathic hepatitis that killed so many infants in the 1970s and 1980s is today, in large part, a gene panel result: PFIC types one to three, tight junction protein two deficiency, MYO5B disease, and the bile acid synthesis defects. Some of these have specific treatments, such as oral cholic acid for bile acid synthesis defects and, for PFIC, the ileal bile acid transporter inhibitors. The persistently acholic group contained missed or late biliary atresia, severe canalicular block such as PFIC type two, and Alagille syndrome with duct paucity. The acute fulminant group contained gestational alloimmune liver disease, tyrosinemia, galactosemia, and herpes or enterovirus infection. What remains as good-prognosis idiopathic hepatitis is the sporadic infant with pigmented stools who grows normally and whose cholestasis resolves by six to twelve months. For counseling, a recurrence in a sibling moves an infant out of the idiopathic category until genetic testing is complete, and the recurrence risk for a recessive cause is one in four.

Part seven. Giant cell hepatitis with autoimmune hemolytic anemia: the phenotype.

Now to the second child from the opening. The best clinical description of the anemia variant comes from Maggiore and colleagues, who gathered sixteen children from Italian and French centers over twenty-eight years. The median age at onset was six months, and across the literature onset ranges from about one month to two years. All sixteen had hepatomegaly, fourteen were jaundiced, nine were pale, eight had fever and six had splenomegaly. The median hemoglobin was six point seven grams per deciliter, and the median alanine aminotransferase was about forty-five times the upper limit of normal. The direct antiglobulin test is positive, typically for immunoglobulin G together with the complement component C3.

Three features deserve emphasis. First, the anemia often comes before the hepatitis. So a toddler with autoimmune hemolysis needs liver tests, and a toddler with hepatitis and anemia needs a direct antiglobulin test. Second, the autoantibody profile is not that of autoimmune hepatitis: the antinuclear, smooth muscle and liver kidney microsomal antibodies that define autoimmune hepatitis types one and two are usually absent. Third, it is rare. Fewer than one hundred cases have been published. The course is relapsing and sometimes fulminant, and most deaths, from sepsis or multi-organ failure, occur within three years of diagnosis.

Part eight. The mechanism: complement on the hepatocyte.

Why is this disease different from autoimmune hepatitis? The most important study is by Whitington and colleagues, published in 2014. They compared liver biopsies from six children with the anemia variant, aged four to fifty-two months, with biopsies from six children with juvenile autoimmune hepatitis. They stained for C5b-9, the terminal membrane attack complex of complement, which punches holes in cell membranes. In all six children with the anemia variant there was high-grade deposition of C5b-9 on hepatocytes throughout the lobule. In autoimmune hepatitis there was not.

The proposed sequence is this. Immunoglobulin G autoantibodies bind red cells and, presumably, hepatocyte antigens. The classical complement pathway is activated, which is why C3 is found on the red cells. The membrane attack complex then forms on hepatocytes, causing necrosis and giant cell transformation. This picture resembles gestational alloimmune liver disease, in which maternal antibodies trigger complement-mediated injury to the fetal liver. It suggests that the anemia variant is a humoral, antibody and complement disease, rather than the T cell disease of classic autoimmune hepatitis.

The clinical courses in that study echoed the mechanism. Only one of the six children remitted on standard autoimmune hepatitis therapy; one died and one needed a transplant. All three who received rituximab, an antibody that depletes B cells, the cells that make antibodies, went into remission. Blocking terminal complement with eculizumab is mechanistically attractive, but

we found no published case or series using it in this disease, so it remains a hypothesis rather than evidence.

Part nine. Treatment and outcome.

In the Maggiore cohort, every child was started on prednisone with azathioprine, and three also received cyclosporine. Eight reached complete remission, six partial remission, and two failed treatment. Relapse was the rule: eleven had a relapse of hepatitis and ten a relapse of anemia, which required prolonged immunosuppression, splenectomy or rituximab. Seven eventually stopped all therapy, after a mean of six years, without relapse. Four of the sixteen died, which is twenty-five percent, from sepsis or multi-organ failure, and one child was alive nine years after liver transplantation. The median follow-up was fourteen years.

Outcomes have improved as rituximab came into use. Poddighe and colleagues grouped published cases by era. Mortality was about forty-five percent in the twenty-two cases reported up to 2006, twenty-five percent in the Maggiore cohort, and about ten percent, three of twenty-nine, in reports after 2011, a period in which sixteen of those twenty-nine children received rituximab. In the Shanghai cohort by Zhang and colleagues, complete remission was reached in seventy-three percent of children on conventional therapy, eleven of fifteen, and in all five children who received rituximab as part of first-line treatment. The median time to remission was six months with rituximab compared with fourteen months with conventional therapy, a statistically significant difference, and the two deaths were both in the conventional group. Across fifty-four treated cases reported since 1981, Nastasio and colleagues estimated a twenty-year survival of sixty-six percent.

The treatment ladder looks like this. First line is prednisone at two to three milligrams per kilogram per day with azathioprine at one to two milligrams per kilogram per day. Cyclosporine or tacrolimus is used as second line or rescue. Rituximab, usually three hundred and seventy-five milligrams per square meter weekly for four doses, with maintenance infusions in severe cases, achieved complete remission in four of four children in a Canadian series by Paganelli, four of four in a Polish series by Bakula, and five of five in the Shanghai cohort. Expert reviews now describe it as the cornerstone for relapsing or complicated disease and an acceptable first-line, steroid-sparing option. Intravenous immunoglobulin lowered aminotransferases and spared steroids in a series by Marsalli, but every one of the seven children relapsed, so its benefit is temporary. Alemtuzumab, an antibody against the CD52 molecule, has been reported as salvage in a single case. Rituximab has a cost: prolonged B cell depletion and low immunoglobulin levels, persistent in two reported children. Monitor immunoglobulin G, replace it if it is low, and review the vaccine schedule.

Liver transplantation deserves a specific warning. Of ten transplanted children with follow-up in the Nastasio review, only two had no recurrence, and an earlier review of four transplants found that three died and all had recurrence in the graft within months. The disease is driven

by the immune system, not by the liver, so a new liver does not cure it. Avoid transplantation if at all possible, and treat early and hard instead.

Part ten. Giant cell hepatitis after infancy.

Giant cell hepatitis in older children and adults is a different question. A review by Bihari and colleagues counted about one hundred reported cases, with autoimmune disease accounting for about forty percent, and noted that viral cases, from paramyxovirus, hepatitis A virus or Epstein-Barr virus, can run a fulminant course. Survival without transplantation was about fifty percent. A 2022 meta-analysis of sixty-eight adults by Jiao and Zhang found an autoimmune cause in thirty-two percent, a viral cause in twenty-one percent, drugs or herbal products in ten percent, malignancy in seven percent, and more than one contributing cause in twenty-four percent. Poor outcome was associated with low platelets, low albumin, high bilirubin and diffuse giant cell change. In an older child with giant cells on biopsy, look for autoimmune hepatitis, a virus or a drug.

Part eleven. What the guidelines say.

The 2017 joint guideline from NASPGHAN and ESPGHAN covers both sides of the Atlantic for the cholestatic infant, so North America and Europe are aligned on three points: idiopathic neonatal hepatitis is a descriptive and declining term, genetic testing should be used with a low threshold, and biopsy is recommended when the evaluation is unclear, with ninety to ninety-five percent accuracy in experienced hands. On ursodeoxycholic acid there is a gap. The guideline mentions it only as a reason to send bile acid studies before starting it, and makes no recommendation about efficacy; we found no randomized evidence for it in idiopathic neonatal hepatitis.

For the anemia variant, the gap is larger. It is not addressed in the joint neonatal guideline, and to our reading it is not covered by the American Association for the Study of Liver Diseases guidance on autoimmune hepatitis, by the 2015 autoimmune hepatitis guideline of the European Association for the Study of the Liver, or by the scope of the 2018 ESPGHAN position statement on pediatric autoimmune liver disease. Indeed, Nastasio and colleagues published a paper in 2018 asking whether it should be considered a pediatric autoimmune liver disease at all. Treatment recommendations therefore come from expert reviews, mainly European, which favor early rituximab, and those reviews advise against transplantation because of early recurrence. We consider this a material gap: no society guideline covers a disease whose mortality was twenty-five to forty-five percent before rituximab.

Part twelve. Traps.

Four traps recur. The first is a report that says neonatal hepatitis in an infant whose stools are pale: an early biopsy can miss atresia, and stool color overrules histology. The second is a label

that never expires: idiopathic neonatal hepatitis written at six weeks and still in the problem list at six months, with no gene panel ever sent, until a sibling is born with the same disease. The third is starting ursodeoxycholic acid before a urine bile acid profile, which can mask a bile acid synthesis defect and delay oral cholic acid, a specific and effective therapy. The fourth is the anemic toddler whose hemolysis is treated alone, with no liver tests, until the anemia variant is discovered at the point of liver failure. And a fifth: transplanting a child with the anemia variant as if it were end-stage autoimmune hepatitis, only to see the disease return in the graft.

Part thirteen. What to do differently in clinic.

Let us return to the two children from the beginning and turn the logic into decisions. For the infant with giant cell hepatitis on biopsy, the sequence is to earn the idiopathic label rather than assume it. First, exclude biliary atresia, which is time-critical: look at the stools yourself, review the ultrasound, have an experienced pathologist read the biopsy, and go to cholangiography if any doubt remains. Second, exclude treatable causes: sepsis and urinary infection, galactosemia and tyrosinemia, cortisol and thyroid function, cytomegalovirus, the alpha-one antitrypsin phenotype, and a sweat test. Third, send a urine bile acid profile before starting ursodeoxycholic acid. Fourth, send genetics early, with a cholestasis panel or a trio exome, and do so at once if the gamma-glutamyl transferase is low, the parents are related or a sibling was affected. Fifth, support the infant with formula enriched in medium chain triglycerides, fat-soluble vitamins with monitoring, and vitamin K, and revisit the label at every visit. Re-label when a gene result returns, when a sibling is affected, when cholestasis persists beyond about six months, or when Coombs-positive hemolysis appears after the newborn period.

Some concrete scenarios make this sharper. A seven-week-old with giant cell hepatitis and pale yellow stools needs cholangiography now, because the biopsy does not exclude atresia. A ten-week-old with pigmented stools, a normal gamma-glutamyl transferase and an idiopathic label has a low gamma-glutamyl transferase pattern and needs a cholestasis panel and urine bile acids before any ursodeoxycholic acid. A second sibling with idiopathic neonatal hepatitis should be treated as having a recessive genetic disease, with trio exome and counseling about a one in four recurrence risk. An infant with giant cell hepatitis, hypoglycemia and micropenis needs cortisol, free thyroxine, thyroid stimulating hormone and pituitary imaging, and hormone replacement. An infant whose idiopathic cholestasis resolved by six months and who is growing well can be reassured, with liver tests and vitamin levels rechecked, and genetics if anything recurs.

And for the nine-month-old with a hemoglobin of six, a direct antiglobulin test positive for immunoglobulin G and C3, and an alanine aminotransferase of one thousand eight hundred: this is the anemia variant until proven otherwise. Start prednisone with azathioprine, bring in

rituximab early rather than as a last resort, check and follow immunoglobulin G, and avoid transplantation if you possibly can.

The single sentence to carry away is this. Idiopathic neonatal hepatitis is not a diagnosis but a to-do list; giant cells describe the liver's response, not its cause; and when a toddler has both hemolysis and hepatitis, think of complement, think of rituximab, and think twice before transplant.